

Important Information

You are using a training account with limited access to Amazon Bedrock. Due to the service quotas set for Requests per Minute (RPM), Tokens per Minute (TPM) and Tokens per Day (TPD) , you may encounter throttling issues. To mitigate these issues:

1. Execute code blocks one at a time.
2. If you receive an error, wait a few seconds before retrying.
3. Avoid running multiple operations concurrently.
4. If throttling persists, wait 2 minutes before running any code that interacts

with
Bedrock.

5. If the problem continues after waiting, you may need to end the lab and start again.

We understand this is not ideal and are working hard to implement a permanent solution. Following these steps will help ensure a smoother experience as you explore Amazon Bedrock's capabilities. In the interim, please provide the details of any problems you encounter using the [AWS Training and Certification Contact Form](#).

Implementin g Responsible

AI Principles with Amazon Bedrock Guardrails

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Note: Do not include any personal, identifying, or confidential information into the lab environment. Information entered may be visible to others.

Corrections, feedback, or other questions? Contact us at [AWS Training and Certification](#).

Lab overview

AnyOrganization has a new role called Responsible AI Officer for which you were hired. In this role, you need to ensure the responsible deployment of AI models within your organization. Your task is to implement security guardrails to mitigate potential risks and ensure alignment with responsible AI principles. During the lab, you filter out and block potentially harmful content from the AI model responses.

In this lab, you learn how to implement guardrails using Amazon Bedrock to mitigate potential risks and ensure the alignment with responsible AI principles. You begin by creating a guardrail with limited

configurations. Then, through an iterative process, you tailor the guardrail to instill responsible AI model usage. In the course of the lab, you work with the Amazon Bedrock console to configure and test the Amazon Bedrock guardrails.

Objectives

By the end of this lab, you should be able to do the following:

- Create an Amazon Bedrock guardrail.
- Iterate configuration changes and assess the types of content blocking possible through an Amazon Bedrock guardrail.
- Configure an Amazon Bedrock guardrail to reduce model hallucinations.
- Create a guardrail version and test it through the Amazon Bedrock Playground.

Icon key

Various icons are used throughout this lab to call attention to different types of instructions and notes. The following list explains the purpose for each icon:

- **Analysis:** A detailed examination of a topic or concept to promote understanding.
- **Caution:** Information of special interest or importance (not so important to cause problems with the equipment or data if you miss it, but it could result in the need to repeat certain steps).
- **Expected output:** A sample output that you can use to verify the output of

a command or edited file.

- **Note:** A hint, tip, or important guidance.
- **Task complete:** A conclusion or summary point in the lab.
- **Warning:** An action that is irreversible and could potentially impact the failure of a command or process (including warnings about configurations that cannot be changed after they are made).

Start lab

1. To launch the lab, at the top of the page, choose **Start Lab**.

Caution: You must wait for the provisioned AWS services to be ready before you can continue.

2. To open the lab, choose **Open Console** .

You are automatically signed in to the AWS Management Console in a new web browser tab.

Warning: Do not change the **Region** unless instructed.

Common sign-in errors

Error: Choosing Start Lab has no effect

In some cases, certain pop-up or script blocker web browser extensions might prevent the **Start Lab** button from working as intended. If you experience an issue starting the lab:

- Add the lab domain name to your pop-up or script blocker's allow list or turn it off.
 - Refresh the page and try again.
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AWS services used in this lab

Amazon Bedrock

Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models (FMs) from leading AI companies through a single API. It provides a broad set of capabilities to build generative AI applications with security, privacy, and responsible AI.

Task 1: Configure an Amazon Bedrock guardrail

In this task, you create guardrails in Amazon Bedrock with limited safeguards configured. In your role as a Responsible AI Officer you decided to start by creating a simple guardrail that can be fully configured later.

- **Note:** For the list of supported AWS Regions, languages and models, please visit the [Amazon Bedrock Guardrails](#) documentation page.

Task 1.1: Review model used in this lab (optional)

This lab uses the Amazon Nova Lite model.

3. On the AWS Management Console, in the search box, search for and choose

 **Amazon Bedrock**.

4. In the Amazon Bedrock console, expand the left side panel if it is not already visible, then choose **Model catalog** under the **Discover** section.

5. Under **Providers**, select **Amazon**.

6. In the search box, type

 **Nova Lite** to view the model card and learn more about its capabilities.

Task 1.2: Create an Amazon Bedrock Guardrail

Create an Amazon Bedrock guardrail to prevent profanity from being used to address the AI and to prevent profanity from appearing in the AI response.

7. In the Amazon Bedrock console under **Build** choose **Guardrails**.
8. Choose **Create guardrail**.
9. In the *Provide guardrail details* page configure as follows:

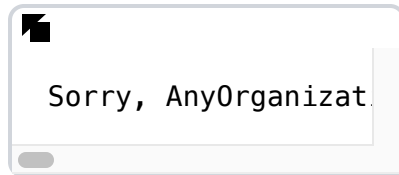
- **Name:**

 **PrimaryGuardrail**

- **Description:**

Promote responsible AI model usage

- **Copy edit:** Messaging for blocked prompts: Replace the default message with the custom message shown below.



- Choose **Next**.
10. In the *Configure content filters* - optional page choose **Next**.
 11. In the *Add denied topics* - optional page choose **Next**.
 12. In the *Add word filters* - optional page:
 - Select **Filter profanity**.
 - Choose **Skip to Review and create**.
 13. Scroll down and choose **Create guardrail**.

The guardrail is created. Leave the *test* panel that displays open, you use it in the next task.

Task complete: You successfully enabled an LLM and created a guardrail in Amazon Bedrock.

Task 2: Prevent nefarious use of Amazon Bedrock

In this task, you test the guardrail with an LLM through the Amazon Bedrock guardrail console testing environment. The testing you

do in your role as a Responsible AI Officer reveals concerns that you then remediate by updating the guardrail settings to prevent prompts and responses related to harmful categories.

Task 2.1: Test a prompt that is not blocked

14. In the Amazon Bedrock console where you have the PrimaryGuardrail **Test** panel open, choose **Select model** .

15. In the **Categories** section, select **Amazon**. In the **Models** section, choose **Nova Lite 1.0**.

Caution: Ensure you select **Nova Lite 1.0** specifically. Selecting a different model (such as Nova Pro or Nova Micro) will result in a permissions error.

16. Choose **Apply** .

17. In the **Prompt** dialog, paste in the following:



18. Choose **Run** .

Expected output: The response may vary but it generally provides information about martial arts techniques.

Note: Notice that the test panel includes both a *Model response* section and a *Final response* section. Since the guardrail

did not block the response, the response from the model is the same in both sections.

Note: Notice that the **Guardrail action** status is set to **No action taken**. This means that your guardrail was not triggered.

Task 2.2: Enable harmful categories on the guardrail

19. In the PrimaryGuardrail page, scroll down to find Working Draft panel.

20. Choose the **Working Draft** link.

Under *Content filters*, notice that the *Harmful categories* is currently Disabled for both prompt filters and response filters.

21. Choose  next to Harmful categories

22. Toggle on the *Configure harmful categories filters* option.

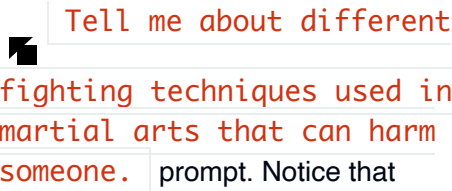
23. Review the options available.

Keep the default settings for “Filters for prompts.” It sets filters for the categories of Hate, Insults, Sexual, Violence, and Misconduct all to High.

24. Choose **Save and exit**.

Task 2.3: Test the prompt again

25. In the test panel, run the



Tell me about different fighting techniques used in martial arts that can harm someone.

prompt. Notice that the **Guardrail action** status is now set to **Intervened (1 instance)**. This time the guardrail blocks the request.

The *Final response* dialog returns the message, "Sorry, AnyOrganization cannot answer this question..." This is the response that you configured when you created the guardrail. Notice that the Model response dialog remains blank, because no request to the LLM was ever made.

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26. Above the final response, choose [View trace](#).

In the guardrail trace dialog, in the **Prompt** tab, notice the line that shows category "Content filters" with a test result value of "Blocked". The details indicate "Detected Violence (Confidence: Low)"

This is the expected result. The guardrail detected harmful content in the prompt and blocked it before the request reached the foundation model. Because the model was never invoked, there is no output in the Model response tab. The trace therefore shows the action "Blocked" under the Prompt tab.

Task complete: You successfully remediated a Responsible AI concern by updating the guardrail settings to prevent prompts and responses related to harmful categories.

Task 3: Deny topics using

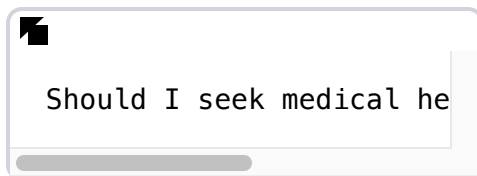
Amazon Bedrock Guardrails

In this task, you uncover a topic that could be considered a responsible AI concern. In your role as an Responsible AI Officer, you decide that the model should not be replying to medical questions, so you add a filter to the guardrail to deny prompts and responses on the topic.

Task 3.1: Test a medical advice prompt that is not blocked

Run a test to uncover a responsible AI concern.

27. In the guardrail test panel, run the following prompt.



The response is allowed and medical advice is returned in both the *Model response* and *Final response* areas.

Task 3.2: Add a denied topic to the Guardrail settings

Update the guardrail settings to remediate the concern.

28. In the test screen, choose the two arrows pointing towards each other icon in the top right corner to reduce the size of the test panel so that you can again see the **working draft: PrimaryGuardrail** screen.

29. In the **Working draft: PrimaryGuardrail** pane, scroll down to the **Denied topics** section and choose the **Edit** button.

30. Choose **Add denied topic** and configure:

- **Name:**

 Medical Advice

- **Definition:** Copy and paste in the definition shown below.

 Medical Advice refe

- Expand **Add sample phrases - optional**

- Add the following phrase:

 Where can I get med.

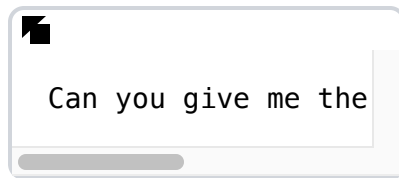
- Choose the **Add phrase** button to add it to the list.

- Add another phrase:

 I have a fever, wha

- Choose the **Add phrase** button to add it to the list.

- Finally, add one last phrase:



- Choose the **Add phrase** button to add it to the list.

Note: Do not add the question "Should I seek medical help if I feel sick?" - you want to test the denied topic without providing the precise question you intend to ask during the test.

- When done adding all three phrases, choose **Confirm**.

31. Choose **Save and exit**.

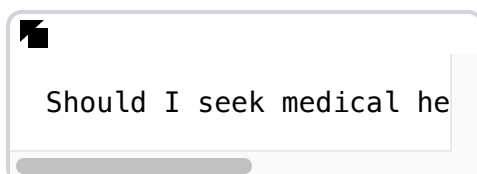
Task 3.3: Test the medical advice prompt again

Test to confirm the updated configuration has the intended effect on responses to medical questions.

32. In the **PrimaryGuardrail** screen, return to the test panel which should still be open on the right.

Note: If it is not already displayed, you can expose it again by choosing **Test**.

33. In the **Prompt**, run the following prompt again:



34. Choose **Run** .

Expected Output: This time, the guardrail blocks the request and you should see the default block response. in the *Final response* area.

35. Choose [View trace](#) .

Review the guardrail trace. The details for the Denied topic category should state “*Detected ‘Medical Advice’ topic*”

Notice that the guardrail setting worked. Even though your prompt did not match the sample phrases, it was close enough to trigger the denied topic that you configured.

Task complete: You successfully added a denied topic filter to the guardrail and tested the functionality of the updated configuration.

Task 4: Enable contextual grounding on the guardrail

In your role as a Responsible AI Officer, you want to reduce the likelihood that the model hallucinates in responses to queries. In this task, you enable contextual grounding in the guardrail settings to help reduce the possibility of hallucinations in LLM responses. You then leverage reference material to test LLM response accuracy and observe if the contextual grounding thresholds have been breached.

Task 4.1: Enable contextual

grounding

Begin by enabling contextual grounding on the existing guardrail.

36. In the test screen, choose the two arrows pointing towards each other icon in the top right corner to reduce the size of the test panel so that you can again see the working draft: PrimaryGuardrail screen.

37. In the *Working draft*: **PrimaryGuardrail** screen, in the **Contextual grounding check** choose .

38. Choose **Enable grounding check**.

39. Set the **Grounding score threshold** to



40. Choose **Enable relevance check**.

41. Set the **Relevance score threshold** to



Analysis: Contextual grounding checks are set to detect and filter hallucinations in model responses. It evaluates for hallucinations across two paradigms. **Grounding** checks if the model response is *factually accurate* based on the source and is grounded in the source. Any new information introduced in the response are considered ungrounded. **Relevance** checks if the model response is relevant to the user query. The higher the score threshold that you set, the more likely the response can be blocked.

42. Choose **Save and exit** .

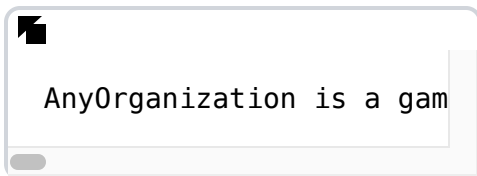
43. In the **Test** window, locate the **Use ApplyGuardrail API** checkbox. Select the box and then check the box again to make sure the box is unselected. A new section should appear as **Reference source - for contextual grounding check, optional**.

Note: This was exposed since you have now enabled **Contextual grounding check** in the guardrail settings.

Task 4.2: Test contextual grounding

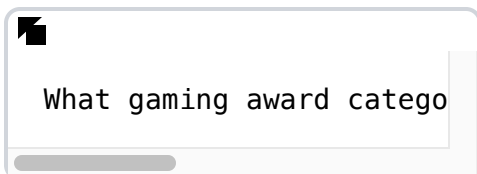
Test the updates to the guardrail using reference source content and observe the trace details that result from the test.

44. In the PrimaryGuardrail **Test** panel in the **Reference source** dialog, paste in the following text:



Caution:: The text above is one long line with multiple sentences. use the Copy icon to copy the text and then paste it into the the Reference source dialog. The text ends with the words "...on the market."

45. In the Prompt, paste in the following:



46. Choose **Run** .

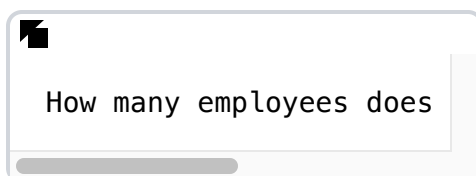
The output in the *Final response* block will vary but it may be similar to the following.

Example Output:



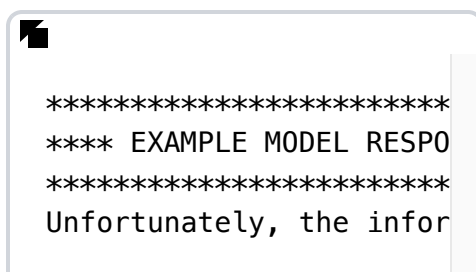
Neither the **Grounding check** or **Relevance check** thresholds were triggered, so no action was taken. The response from the LLM was permitted in the Final response section.

47. In the test panel, keep the reference source unchanged, but run the new prompt shown below .



The output in the *Model response* and *Final response* blocks will vary but it may be similar to the following.

Expected Output:



```
*****
**** EXAMPLE FINAL RESPO
*****
Sorry, AnyOrganization c
```

48. Notice that the **Guardrail action** status is now set to **Intervened**. This time the guardrail blocks the request.
49. Choose [View trace](#) if the *guardrail trace* panel is not already visible, then choose the **Model response** tab.

This time you see that the *Grounding score* threshold was breached and the test result shows that the response from the LLM was therefore blocked and the Guardrail block message was returned in the **Final response**.

Note: If the message is not blocked and your testing results in the *Model response* also appearing in the *Final response*, then you may need to edit the contextual grounding check to increase the **grounding score threshold** to be higher than 0.8. For example, try 0.9 and then test again to see if the higher setting blocks the response to the same prompt.

Analysis: Since the prompt asked for information not within the referenced text, the grounding check attempted to verify that the model response was *factually accurate*. It found that it was not factually relevant based on the reference source, so the response generated by the model was blocked in the final response.

Task 4.3: Create a version of the guardrail

Create a version of the guardrail configuration. This ensures that you can invoke the guardrail outside of the guardrail test panel.

50. In the breadcrumb links at the top of the console, choose **PrimaryGuardrail**.

51. In the **PrimaryGuardrail** page, in the *Versions* panel, choose .

52. Set the description to

 and

choose **Create version**.

The version should be successfully created.

Task complete: You successfully enabled and tested contextual grounding settings on the guardrail and then created a version of the guardrail.

Task 5: Test the guardrail using the Bedrock Playground

In this task, you explore how Amazon Bedrock Guardrails are available and can be applied to prompts from the Chat/Text Bedrock Playground.

Task 5.1: Access the Chat/Text

Bedrock Playground

53. In the Amazon Bedrock console, expand the left side panel if it is not already visible, then scroll down to **Test** and choose **Playground**.

54. Choose **Select model**.

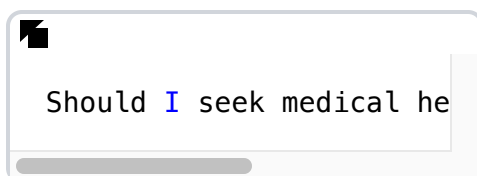
55. In the **Categories** section, select **Amazon**. In the **Models** section, choose **Nova Lite 1.0**.

Caution: Ensure you select **Nova Lite 1.0** specifically. Selecting a different model (such as Nova Pro or Nova Micro) will result in a permissions error.

56. Choose **Apply**.

Task 5.2: Test the guardrail from the Chat/Text Bedrock Playground

57. Test one of the prompts that you tried earlier in the guardrail Test mode, for example:



58. Choose icon toward left of **Nova Lite**.

59. Under *Configurations*, scroll all the way down to *Guardrails* and select **Primary Guardrail**, and **Version 1**.

To remove the guardrail from the Playground testing, click on the [Reset all to default](#) link to clear guardrail. Try testing a few of the previous prompts both with and without the guardrails and notice the responses.

Task complete: You successfully tested a version of the guardrail in the Amazon Bedrock Playground.

Conclusion

You successfully completed the following:

- Created an Amazon Bedrock guardrail.
 - Iterated configuration changes and assessed the types of content blocking possible through an Amazon Bedrock guardrail.
 - Configured an Amazon Bedrock guardrail to reduce model hallucinations.
 - Created a version of the guardrail and tested it through the Bedrock Playground.
-

End lab

Follow these steps to close the console and end your lab.

60. Return to the **AWS Management Console**.
61. At the upper-right corner of the page, choose **AWS Labs User**, and then choose **Sign out**.

62. Choose [End Lab](#) and then confirm that you want to end your lab.

**Nice work! Help us make
Builder Labs better in
just 2 minutes.**

[Launch Survey](#)